21-242 Recitation 12

Fundamental theorems — Q2. T A injective implies A has a left inverse

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that T_A injective implies A has a left inverse.

Course-level solution

Let $A \in \mathbb{F}^{m\times n}$ with $T_A : \mathbb{F}^n \to \mathbb{F}^m$ injective. Then $\ker T_A = \{\vec{0}\}$, so the columns of A are independent. In particular $n \le m$.
Let $\mathcal{B} = (Ae_1, \dots, Ae_n)$, a linearly independent list in $\mathbb{F}^m$. Extend it to a basis $\mathcal{C} = (\vec{w}_1, \dots, \vec{w}_m)$ of $\mathbb{F}^m$ with $\vec{w}_j = Ae_j$ for $j \le n$ (possible because we may add vectors from a spanning list, e.g. the standard basis, that are not yet in the span).
Define $S : \mathbb{F}^m \to \mathbb{F}^n$ to be the unique linear map with

$$S(\vec{w}_j) = \begin{cases} e_j & 1 \le j \le n, \\ \vec{0} & n < j \le m. \end{cases}$$

Then $S(T_A(e_j)) = S(\vec{w}_j) = e_j$ for $j = 1, \dots, n$, so $S \circ T_A = \mathrm{id}_{\mathbb{F}^n}$. Letting B be the standard matrix of S, we have $BA = I_n$.

Later-course solution (using ideas from the rest of the handout)

By rank-nullity, injectivity means $\mathrm{rank}(A) = n$. Then $\mathrm{RREF}(A) = \begin{pmatrix} I_n \\ 0 \end{pmatrix} = GA$ for invertible G, so the first n rows of G form a left inverse of A. This is the computational version of the same fact.